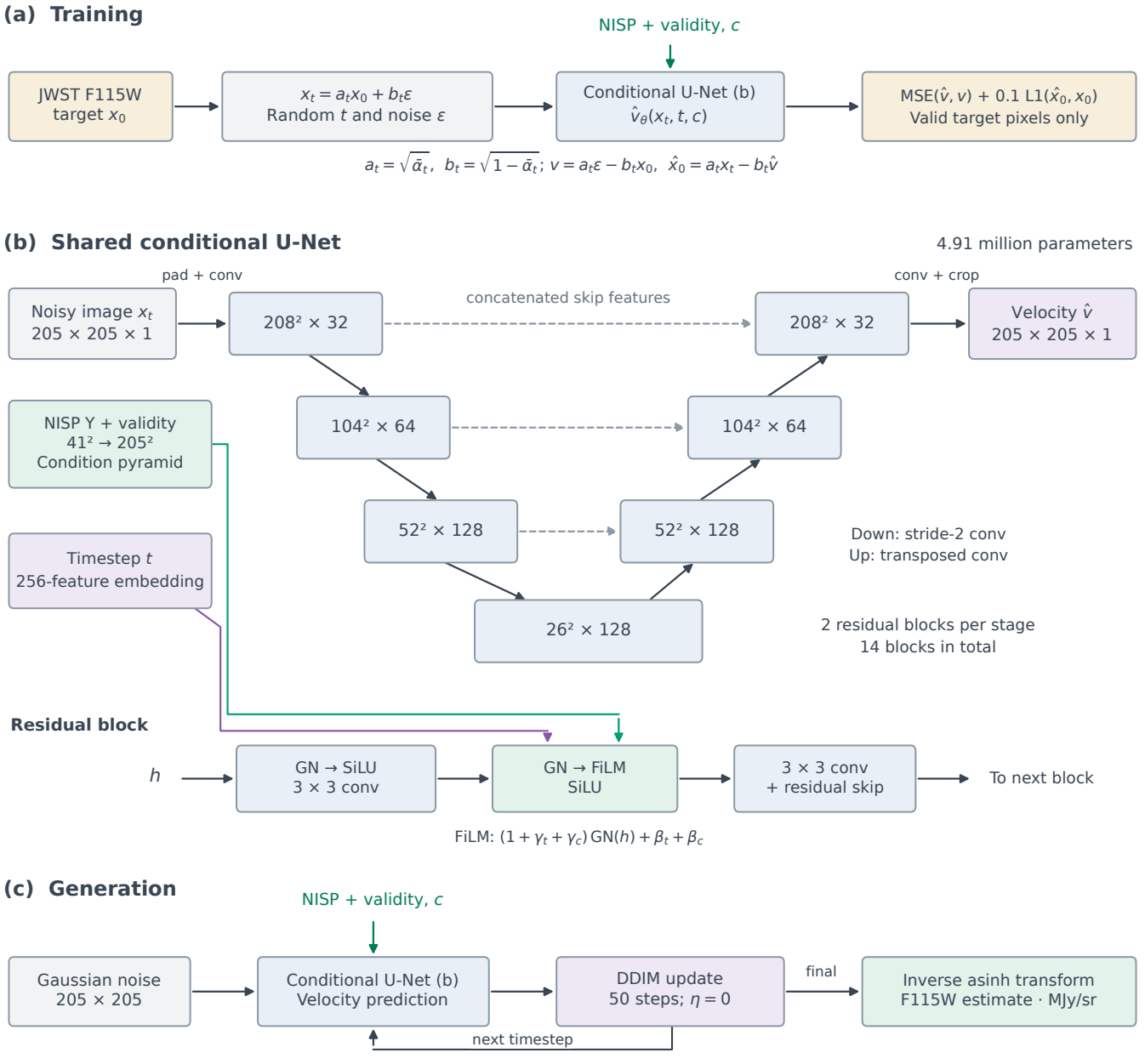
(a) Training
NISP + validity, c
JWST F115W target x0
xt = at x0 + bt ε
Random t and noise ε
Conditional U-Net (b)
v̂θ(xt, t, c)
MSE(v̂, v) + 0.1 L1(x̂0, x0)
Valid target pixels only
at = √ᾱt, bt = √1 − ᾱt; v = at ε − bt x0, x̂0 = at xt − bt v̂
(b) Shared conditional U-Net
4.91 million parameters
pad + conv
conv + crop
Noisy image xt
205 × 205 × 1
208² × 32
concatenated skip features
208² × 32
Velocity v̂
205 × 205 × 1
NISP Y + validity
41² → 205²
Condition pyramid
104² × 64
104² × 64
Timestep t
256-feature embedding
52² × 128
52² × 128
Down: stride-2 conv
Up: transposed conv
26² × 128
2 residual blocks per stage
14 blocks in total
Residual block
h
GN → SiLU
3 × 3 conv
GN → FiLM
SiLU
3 × 3 conv
+ residual skip
To next block
FiLM: (1 + γt + γc) GN(h) + βt + βc
(c) Generation
NISP + validity, c
Gaussian noise
205 × 205
Conditional U-Net (b)
Velocity prediction
DDIM update
50 steps; η = 0
final
Inverse asinh transform
F115W estimate · MJy/sr
next timestep